

Statement of Work
High-Power, Frequency-Doubled Blue Laser System for Calcium
Clock Development Division, Precise Time Department
U. S. Naval Observatory

BACKGROUND

The United States Naval Observatory (USNO) is the authoritative source of time for the Department of Defense. For this purpose, the USNO's Precise Time Department maintains an ensemble of high-performing atomic clocks that is used to generate precise time. Because of the lack of commercial vendors for the highest performing clocks, stocking the clock ensemble with systems based on cutting-edge technology requires that these systems be designed and built in-house.

Within the Precise Time Department, the Clock Development Division is investigating atomic clock technology using optical spectroscopy on an atomic beam of calcium. For research and development efforts, USNO requires a high-power, frequency-doubled laser centered at 423 nanometers (nm) with a fundamental at 846 nm to allow us to phase lock to a user provided optical reference. Additionally, high-bandwidth laser locking electronics are required to narrow the laser linewidth below the free-running linewidth of the laser system.

SCOPE

The USNO is seeking one high power 423nm laser system and one digital laser locking device. The full list of requirements is listed in the next section.

REQUIREMENTS:

1. System needs to use an amplified 846 nm source laser that frequency doubles to a wavelength centered around 423 nm
2. Linewidth: 10kHz (kilohertz) or less
3. Mode-hop free tuning range about required wavelength: 15GHz (gigahertz)
4. Generated 423 nm output must be free space with at least 1000 mW (milliwatts) exiting the laser.
5. The 846 nm light must be generated by an external cavity diode laser (ECDL) and tapered amplifier chip (TA) configuration
6. Optical isolation: System must include optical isolation between the 846 nm source and subsequent amplifier and between the amplifier and the frequency doubler.
7. System must include a pick-off to be able to measure the wavelength of the source laser at 846 nm. This light must be fiber coupled.
8. System must include auto-tuning of the alignment of the source laser into the tapered amplifier and the amplified output into the second harmonic generation (SHG) stage.
9. System must include auto-relocking of the resonant bow-tie cavity used for frequency doubling, preferably using the Pound-Drever-Hall method of stabilization.
10. The system must include an electro-optic modulator for the SHG stage to remove residual

frequency or amplitude modulation

11. The final spatial mode out of the frequency doubler must be well shaped and collimated.
12. The system must be “turn-key” and able to achieve these specifications in a standard laboratory environment without user-supplied parts of assembly.

Additionally, the vendor needs to provide a laser locking system that seamlessly interfaces with the laser controller and meets the following requirements:

13. Capable of locking to high finesse cavities or frequency combs and phase locking between lasers.
14. Bandwidth: 50 MHz (megahertz)
15. Linewidth: <1 Hz (hertz)
16. PID regulator with 10ns (nanoseconds) or less signal delay

FACILITIES, EQUIPMENT, AND TRAVEL:

Planning and support for the completion of these tasks will be coordinated with the technical POCs listed below for the USNO. No travel is anticipated for this order.

PERIOD OF PERFORMANCE:

The period of performance is five (5) months from the award date